1. Item 7: Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

(Operate fluently with numerical and geometric forms)

- Highly relevant: Students must:
  - Extract dimensions from orthographic projections (front/side views)
  - Recognize that the roof consists of two rectangles
  - Compute the slant height using the Pythagorean theorem: $\sqrt{(1.25)^2 + (0.10)^2}$
  - Multiply to find area: $2 \times 6.00 \times \text{slant height}$
- This involves geometric reasoning, decimal arithmetic, and spatial-to-numerical translation—all central to Item 7.

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2. Item 1: Perform a simple calculation

(Carry out basic arithmetic operations)

- Highly relevant: Although the context is complex, the core computation—once the slant height is found—is straightforward multiplication and addition.
- Even the square root can be approximated or calculated directly (with calculator, if allowed). In many curricula, this is considered a "routine" multi-step calculation.
- Thus, Item 1 applies strongly to the execution phase.

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3. Item 3: Implement a given strategy to determine a mathematical solution

(Follow a clear procedure to solve the problem)

- Moderately to highly relevant: The problem implies a standard strategy:
  1. Identify roof as two rectangles
  2. Determine the dimensions of one rectangle (length = 6.00 m; width = slant height)
  3. Use front view to find rise (0.10 m) and run (1.25 m)
  4. Apply Pythagoras
  5. Compute total area
- While not explicitly spelled out, this is a conventional approach in geometry problems involving gable roofs. Students are expected to implement this known procedure.

4. Item 5: Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

(Apply standard mathematical rules and algorithms)

- Moderately relevant: The use of the Pythagorean theorem is a canonical algorithm for finding diagonal lengths in right triangles.
- Recognizing the roof cross-section as a right triangle and applying this rule demonstrates understanding of a formal geometric construct.

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5. Item 10: Make sense of, relate and use a variety of representations when interacting with a problem

(Interpret and coordinate multiple representations)

- Moderately relevant: Students must integrate:
  - Front view (shows width and roof pitch)
  - Side view (shows depth/length)
  - Textual description ("two identical rectangular sections")
- Synthesizing these into a coherent 3D mental model is essential—this is precisely what Item 10 describes.

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6. Item 11: Switch between different representations in the process of finding solutions

(Translate among representations during problem solving)

- Weakly relevant: The student moves from 2D diagrams → 3D mental image → geometric model (right triangle) → algebraic formula → numerical answer.
- This chain involves representational shifts, though they are typical in geometry tasks and not unusually complex here.

1. 第7条：Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

（熟练操作数字、图形、统计信息、代数表达式与几何表征）

- 高度相关：学生必须：
  - 从正视图与侧视图中提取关键尺寸；
  - 识别屋顶由两个矩形构成；
  - 利用勾股定理计算斜高：$\sqrt{(1.25)^2 + (0.10)^2}$；
  - 进行乘法运算得出总面积。
- 整个过程融合了几何推理、小数运算与空间到数值的转换，正是第7条所强调的核心能力。

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2. 第1条：Perform a simple calculation（进行简单计算）

- 高度相关：尽管情境较复杂，但一旦确定斜高，后续计算（乘法与加法）属于基础算术操作。
- 即使涉及平方根，在允许使用计算器的情况下，这仍被视为一种常规的多步计算。
- 因此，第1条在执行阶段具有很强的相关性。

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3. 第3条：Implement a given strategy to determine a mathematical solution（执行给定策略以得出数学解）

- 中高相关：题目隐含了一个标准解题策略：
  1. 确认屋顶由两块矩形组成；
  2. 确定每块矩形的尺寸（长 = 6.00 米，宽 = 斜高）；
  3. 从正视图获取垂直升高（0.10 米）和水平跨度（1.25 米）；
  4. 应用勾股定理；
  5. 计算总面积。
- 虽然未明说，但这是处理双坡屋顶问题的常规方法，学生需按此既定程序操作。

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4. 第5条：Understand and utilise constructs based on definitions, rules and formal systems...（理解并运用基于定义、规则和形式系统的数学结构，如使用熟悉算法）

- 中等相关：使用勾股定理是求解直角三角形斜边的标准算法。
- 学生需识别屋顶横截面为直角三角形，并正确应用该定理，体现了对正式几何规则的理解。

5. 第10条：Make sense of, relate and use a variety of representations when interacting with a problem（理解、关联并使用多种表征方式）

- 中等相关：学生必须整合：
  - 正视图（显示宽度与屋顶坡度）；
  - 侧视图（显示深度/长度）；
  - 文字描述（“屋顶由两个相同的矩形部分组成”）。
- 将这些信息综合成一致的三维心理模型，正是第10条所描述的能力。

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6. 第11条：Switch between different representations in the process of finding solutions（在解题过程中切换不同表征）

- 弱相关：学生经历了从二维图纸 → 三维想象 → 几何模型（直角三角形） → 代数公式 → 数值答案的转换链。
- 虽然涉及表征切换，但在几何题中属于常规操作，复杂度不高。